

Correlation Between Segmental Dynamics, Glass Transition, and Lithium Ion Conduction in Poly(Methyl Methacrylate)/Ionic Liquid Mixture

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A solid-state membrane of a polymer/ionic liquid miscible mixture, poly(methyl methacrylate) (PMMA) and 1-butyl-3-methylimidazolium hexafluorophosphate (BMIMPF₆) doped with lithium perchlorate (LiClO₄), was prepared and characterized. Miscibility, segmental dynamics, glass transition and ionic conductivity were investigated. Based on the results from differential scanning calorimetry (DSC) and dynamic mechanical thermal analysis (DMTA), the system is fully miscible and of single phase. Broadening of the glass transition was observed when increasing the amount of ionic liquid, which can be attributed to mobility and flexibility differences between the polymer and ionic liquid. A large dynamical asymmetry and intrinsic mobility difference allow segmental and structural motion/relaxation over a wider temperature range by increasing the amount of ionic liquid. Saturation recovery spin–lattice relaxation time (T_1) versus temperature obtained from ⁷Li nuclear magnetic resonance (NMR) showed high mobility of lithium ions, which was almost temperature independent. Lithium ion conductivity significantly increases with increasing ionic liquid amount. It is concluded that lithium ion mobility and its conduction is positively correlated to segmental dynamics of ion carriers in this model system, which is more noticeable in mixtures with higher amounts of the ionic liquid.

Keywords glass transition, lithium ion mobility, segmental dynamics, solid-state electrolytes

Introduction

The increasing demand for energy and the environmental concerns have led to the current waves of search for alternative and renewable energy sources. Electrochemical systems, such as batteries and super capacitors, have efficient energy storage and delivery ability and can provide great power quality and loading in integrated energy systems. In virtue of high energy density, lithium ion batteries are expected to provide the best energy storage among all current energy storage systems. Lithium ion batteries are also considered the best option for sustainable transportation such as hybrid electric vehicles and plug-in electric vehicles.

Received 23 April 2012; accepted 21 July 2012.

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These advantages of lithium ion batteries have triggered recent progress in safety, cycle life, cost, and wide operational temperature range and material availability. It is now commonly accepted that further breakthroughs in lithium battery technology would need innovations in both electrode and electrolyte components.^[1] Electrolyte is a critical component for the performance of high energy density secondary lithium batteries. Lithium ion batteries commonly use liquid electrolytes. These liquid electrolytes are volatile and flammable, therefore leading to a number of safety concerns and severely limiting their overall application.^[2] Recent attempts to overcome the drawbacks of lithium batteries arising from liquid electrolytes have focused on developing solid-state polymer electrolytes.^[3–5] A desirable polymer electrolyte acts as a mechanical separator, an electronic insulator, and an ionic conductor. It should also be electrochemically stable and easily processable.^[6] To address the mentioned safety concerns and to add flexibility of cell design, a great number of studies in the last decade have focused on designing polymer gel and solid electrolytes, meeting the desired criteria.

Recently, there has been an exponential growth in the number of studies on electrolytes consisting of ionic liquids (ILs), organic-based molten salts, which are liquid at ambient temperature.^[7–14] As salts, ILs have a very low vapor pressure and therefore are nonvolatile, nonflammable, and highly ion conductive. An additional advantage of ILs is their excellent solubility for lithium salts, as in 1-butyl-3-methylimidazolium hexafluorophosphate (BMIMPF₆) systems where homogeneous binary BMIMPF₆/lithium salt mixtures are obtainable over a wide range of lithium salt concentrations. The presence of both positively and negatively charged ions also allows achieving high values of ionic conductivity.^[15,16]

Previous studies on IL-based electrolytes have focused on ionic conductivity of gel-type polymer electrolytes.^[15–19] In this case, the IL in the gel system plays the role as a plasticizer and a medium for the short-range conduction of lithium ions. This concept is expected to be helpful for the design of an IL-based gel electrolyte, providing good ionic conductivity. It is noted, however, that these studies deal with electrolytes with high amounts of ILs. Although conductivity is reasonably improved, other problems such as lack of mechanical strength and film-forming and standing ability, leakage, and safety issues are present. One of the main disadvantages with gel-type membranes made of IL/polymer mixtures is the lack of structural stability, which results in changes in ionic conductivity, thermal stability, and electrochemical stability over time.^[1]

Despite a great number of studies on gel-type polymer electrolytes, to the best of our knowledge, there has been no significant research effort on solid-state polymer/IL systems. Ion conduction in solid polymer electrolytes is highly dependent on segmental motions of ion carrier polymer chains.^[6,20–23] In studying conductivity of solid-state polymer electrolytes, an important aspect is to investigate polymer segmental dynamics and motions of the ion carriers. Blending and mixing are convenient methods for preparing new materials with desired properties. Dynamical properties of polymer blends have long been an important topic from both fundamental and technological point of view. Access to tailor-made materials could be greatly improved if the dynamical characteristics of these materials at the molecular or, more specifically, at both segmental and terminal levels were understood.^[24] In the context of dynamical properties, some novel behavior has been reported for miscible polymer blends.^[20,25,26] In the case of miscible blends with a large dynamical asymmetry, that is, a large difference in glass transition temperature (T_g) of the two constituent polymers, the low T_g constituent shows accelerated segmental dynamics.^[25] This is an interesting feature since enhancement of conductivity in solid-state electrolytes would be achieved in a binary blend system in which the ion conductive polymer is mixed with

another polymer. The important point here is the miscibility of the different components in the blend/mixture.^[6]

In the present study, we produced miscible mixtures of a relatively rigid polymer with an ionic (glass-forming) liquid. The combination of properties of the rigid polymer matrix and IL could be ideal for optimization of solid-state polymer electrolytes. The phenomenon observed in miscible polymer blends mentioned above was expected to be reproduced in this polymer/low molecular weight glass former system. The only difference is the less marked dynamical contrast between the two components of the mixture (due to the absence of chain connectivity in the IL^[27]), in comparison to miscible polymer blends.^[28–31]

Recently, fabrication of all solid-state lithium ion batteries has been reported.^[32,33] Although interesting, these batteries appear far from practical application due to limited cycling performance. However, as stated by Armand and Tarascon^[34,35] achieving solid-state polymer electrolytes with all required properties to be used in lithium ion batteries requires a better and comprehensive understanding of the fundamentals of lithium ion dissociation, mobility, and transport. Here, we report a thorough study on miscibility, glass transition, and dynamical mechanical properties of PMMA [poly(methyl methacrylate)]/BMIMPF₆ solid-state systems doped with lithium perchlorate (LiClO₄). We present solid-state nuclear magnetic resonance (NMR) results based on the use of ⁷Li saturation recovery spin–lattice relaxation time (*T*₁) to characterize the local lithium environment and its molecular level mobility. These results provide the first step toward understanding the lithium ion local mobility mechanism and serve as a foundation for further investigation of polymer/IL solid-state electrolytes.

Experimental

Poly methylmethacrylate with *M*_w = 350 kg/mol and polydispersity index <1.1, and LiClO₄ (battery grade) with 99.99% purity were purchased from Aldrich (USA). 1-Butyl-3-methylimidazolium hexafluorophosphate (BMIMPF₆) (purity > 98%) was purchased from Acros Organics (USA). Lithium perchlorate at 5 wt% of the total polymer content was used to improve the ionic conductivity of the mixture/blend samples and also to perform a direct comparison between the conductivity in different samples. PMMA/BMIMPF₆/LiClO₄ mixtures were prepared by dissolving the three components in tetrahydrofuran (THF) and stirring at room temperature for 48 hr. The solvent was then removed by casting on a Teflon dish and drying at 313 K for about 24 hr. For complete removal of the solvent, the samples were vacuum dried at 323 K for another 48 hr. To avoid moisture and oxygen, all processes were carried out in a glove box filled with argon gas. Compositions of all investigated mixtures are summarized in Table 1.

Attenuated total reflection Fourier transform infrared spectroscopy (ATR-FTIR) was conducted using a Tensor 27 FTIR spectrometer (Bruker Optics, Canada) and analyzed to

Table 1

Composition dependence of *T*_g for PMMA/BMIMPF₆/LiClO₄ blend systems at different BMIMPF₆ contents; 5 wt% LiClO₄ was added to each of these blend samples

BMIMPF ₆ (wt%)	0	20	40	50
PMMA (wt%)	100	80	60	50
<i>T</i> _g (°C)	123	87	45	20

evaluate possible interactions between LiClO_4 salt and the other components, and dissociation of the lithium salt in the mixture. Samples films with the thickness of 1 mm were prepared by solution casting and used for the ATR-FTIR analysis. For the pure LiClO_4 , a pellet with the thickness of 1 mm was prepared by pressing the powder and used for this analysis.

Spectra acquisition and evaluation were performed with OPUS 4.2 software using 3D and MAP packages (Bruker Optics, Canada). The spectra were recorded at room temperature over a wavenumber range from 4000 to 600 cm^{-1} at a resolution of 2 cm^{-1} in reflectance mode. Background and sample spectra were averaged over 100 scans.

Differential scanning calorimetry (DSC) measurements were carried out by using a Q-DSC (2000) (TA Instruments Co., USA). The heat flow versus temperature thermographs were taken on the second heating (10 K/min) cycle after quenching from above T_g . Samples were annealed for 3 min at 423 K before quenching, to remove prior thermal history.

Dynamic mechanical thermal analysis (DMTA) was carried out on all samples using a Q-DMA (800) (TA Instruments Co.) by implementing a double cantilever bending mode with both frequency and temperature sweeps. For the temperature sweep mode, a heating rate of 10 K/min in the temperature range between 300 and 400 K was used at the fixed frequency of 1 Hz. A frequency range between 0.01 and 100 Hz was used for the frequency mode test at room temperature and some fixed temperatures above room temperature. Test specimens were prepared by molding in a hot press to be 1 mm thick, 10 mm wide, and 20 mm long.

^7Li measurements were performed using an 11 T Bruker spectrometer, operating at a ^7Li -resonance frequency of 195 MHz. A solid-state probe was used for the measurements, with a $\pi/2$ pulse of 5 ms. Spin-lattice relaxation measurements, T_1 , were performed with an inversion-recovery sequence (π - τ - $\pi/2$). All the spectra represent a summation of 16 scans, and the recycle delay time between each scan was 5 s.

Ionic conductivity of the electrolyte membranes, over the temperature range 25°C–85°C, was measured by alternating current (AC) impedance spectroscopy (VM3, Bio-logic, France) using stainless-steel (SS) cells over the frequency range of 1 Hz to 1 MHz at AC amplitude of 5 mV. The bulk resistance of a polymer electrolyte, R_b can be obtained from the impedance spectrum. Ionic conductivity values were calculated from the following equation:

$$\sigma = \frac{d}{R_b S}, \quad (1)$$

where σ is the ionic conductivity, R_b is the bulk resistance, d is the thickness of the electrolyte membrane, and S is the area of the symmetrical electrode.

Results And Discussion

Our study was focused on miscible mixtures/blends with high content of PMMA, that is, solid-state PMMA/BMIMPF₆ mixtures, as indicated in Table 1. ATR-FTIR was employed to clarify how LiClO_4 interacts and associates with the blend components. The FTIR spectra for pure LiClO_4 , pure PMMA, PMMA/BMIMPF₆, and PMMA/BMIMPF₆/ LiClO_4 blends are shown in Fig. 1. Pure LiClO_4 shows a peak at about 3450 cm^{-1} .^[6] As can be seen, for all blend samples, this peak disappears and new peaks around 900 cm^{-1} in spectra d-f indicate dissociation of the salt in the matrix due to favorable interactions between the Li

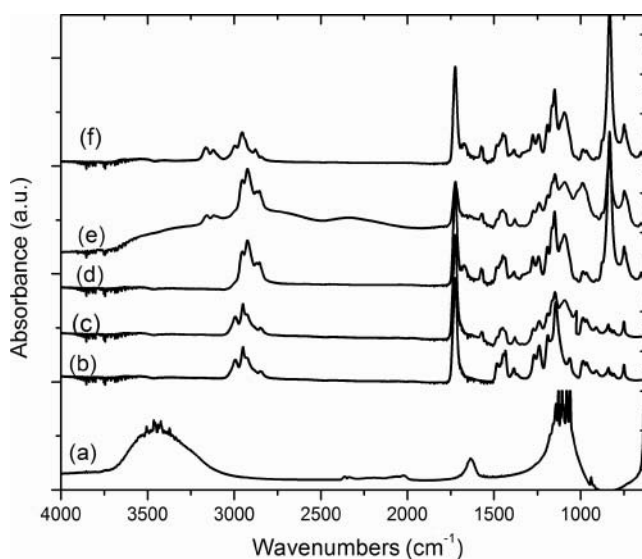


Figure 1. FTIR spectra for (a) LiClO_4 , (b) PMMA, (c) PMMA/BMIMPF₆, (d) PMMA/BMIMPF₆/LiClO₄ (80/20/5), (e) PMMA/BMIMPF₆/LiClO₄ (60/40/5), and (f) PMMA/BMIMPF₆/LiClO₄ (50/50/5).

ions and the IL, and also the PMMA matrix. BMIMPF₆ and LiClO₄ have a high tendency to interact with each other over a wide composition range. The high tendency of Li^+ to associate with the BMIMPF₆ IL has been previously reported.^[36] The tendency of PMMA to interact and dissociate LiClO₄ has also been reported in several studies.^[37–39] Thus, it can be concluded that both of these components, BMIMPF₆ and PMMA, can interact and dissociate the lithium salt. However, it is expected that the main reason for dissociation of the salt is Li^+ interaction with the IL.^[36]

In Fig. 2(a), heat flow versus temperature plots are displayed for pure PMMA, pure BMIMPF₆, and all investigated PMMA/BMIMPF₆/LiClO₄ systems. The corresponding derivative curves are also displayed, in Fig. 2(b). Based on the heat flow thermographs, a single glass transition, although broad, indicates full miscibility and single-phase morphology of the PMMA and BMIMPF₆ in the blend systems. Also, it can be seen that increasing the amount of IL broadens the glass transition range. However, as shown in Fig. 2(b), two distinct glass transition peaks can be observed in derivative curves of heat flow versus temperature for all of the mixture samples. This does not necessarily indicate phase separation and has previously been reported for miscible polymer blends with a large difference in their glass transition temperature due to intrinsic mobility difference between the mixture components.^[31,40,41] In DSC curve, the glass transition temperature of the samples is defined as the middle point of the temperature range in which the heat flow jumps from the value of glassy state to that of rubbery state. Table 1 summarizes the T_g values of all investigated systems as a function of BMIMPF₆ weight fraction.

Figure 3 shows the temperature dependency of storage modulus, loss modulus, and damping factor (tan delta) for the PMMA/BMIMPF₆/LiClO₄ 60/40/5 blend system, as an example. A single damping factor peak indicates full miscibility and homogeneity of this blend system. An outstanding feature in Fig. 3 is the extremely broad tan delta peak,

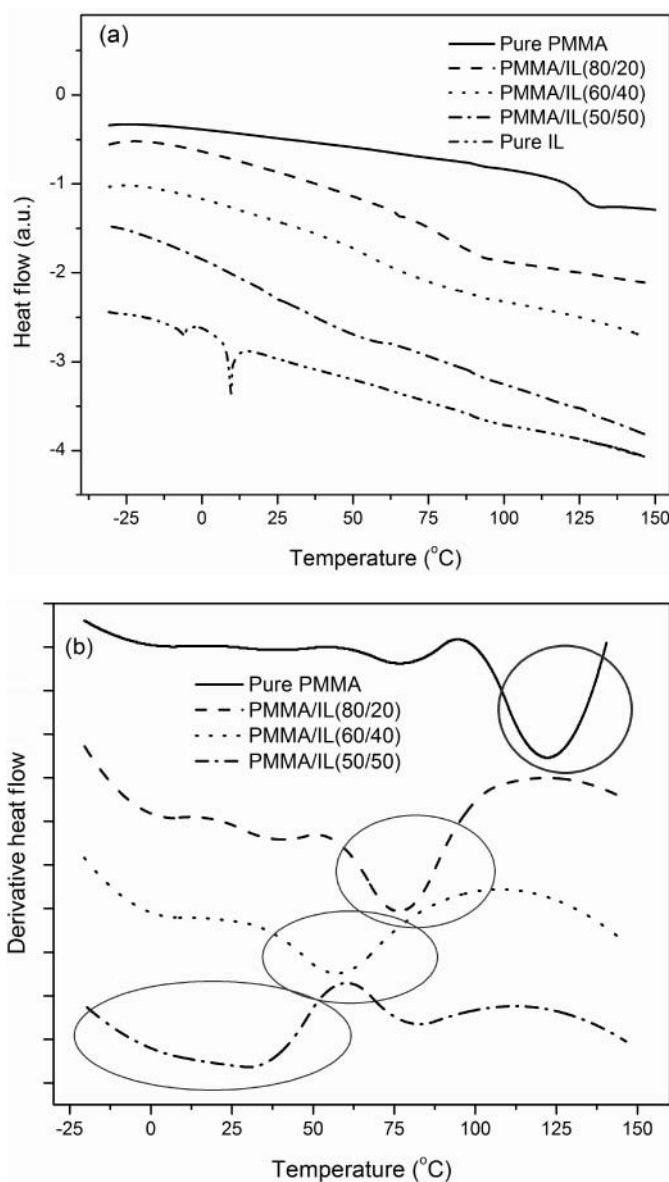


Figure 2. (a) Heat flow versus temperature plots for pure BMIMPF₆, pure PMMA, and PMMA/BMIMPF₆ systems at different blend compositions; (b) the corresponding derivative curves. The ellipses indicate the T_g range.

attributed to relaxation due to the onset of segmental motions of the various constituents in this blend system, starting at low temperatures. This result agrees with that recently found by Simon et al. who also found a significant broadening of the dynamic-mechanical response of a polymer/oligomer mixture.^[42]

Damping factors versus temperature plots for pure PMMA and all the investigated blend samples are shown in Fig. 4. In all samples, the single tan delta peak indicates the complete mixing of the components at the molecular level. As can be seen, an increase in

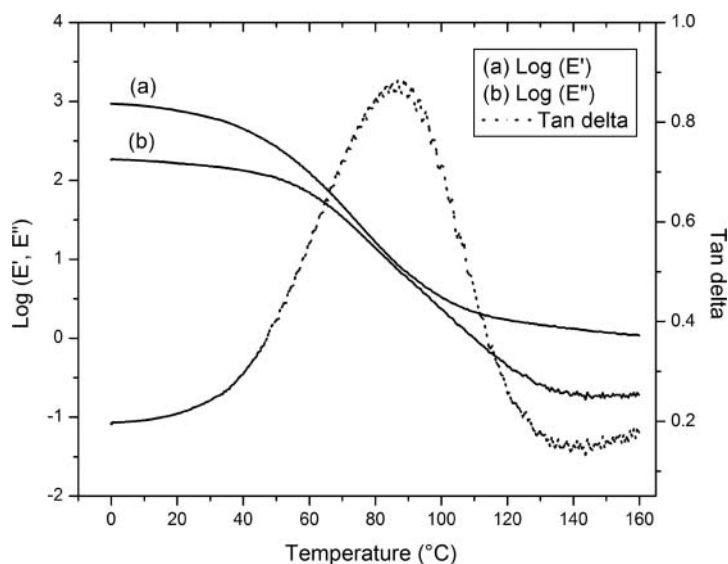


Figure 3. Temperature dependence of storage modulus, loss modulus, and damping factor (tan delta) for PMMA/BMIMPF₆ (60/40) blend system.

IL content shifted the tan delta peak to lower temperature and increased the peak height. This trend is due to the increase in chain flexibility of the system, which leads to relaxation at lower temperature. Since the occurrence of a glass transition is due to cooperative motion of polymer chains at the segmental level, it is expected that the addition of the IL (viscous component) to PMMA (rigid component) results in an increase in damping factor of the system.

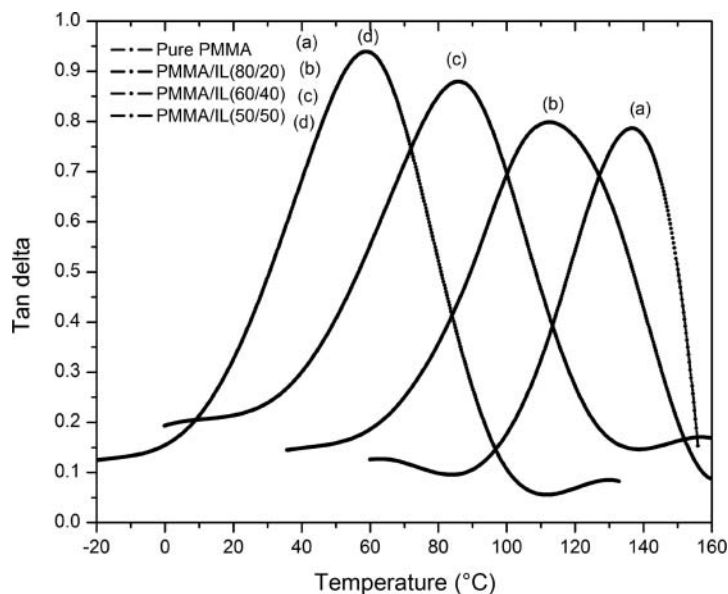


Figure 4. Damping factor versus temperature plots for pure PMMA and all the investigated PMMA/BMIMPF₆ blend samples.

A typical frequency dependence of the measured loss modulus at several temperatures in the glass transition region is shown for the 80/20 blend sample in Fig. 5(a). With increasing temperature, the intensity of the maximum damping peak increased and this peak shifted to slightly higher frequencies. Interestingly, at high temperatures, two distinct peaks were observed. The miscibility of ILs with polymers has been previously investigated,^[43] and the ability of ILs to act as plasticizers has been highlighted in some cases, especially in PMMA.^[44] In fact, an outstanding miscibility of imidazolium salts with methyl methacrylate and its polymers, including the polymer/IL mixture studied here, has been

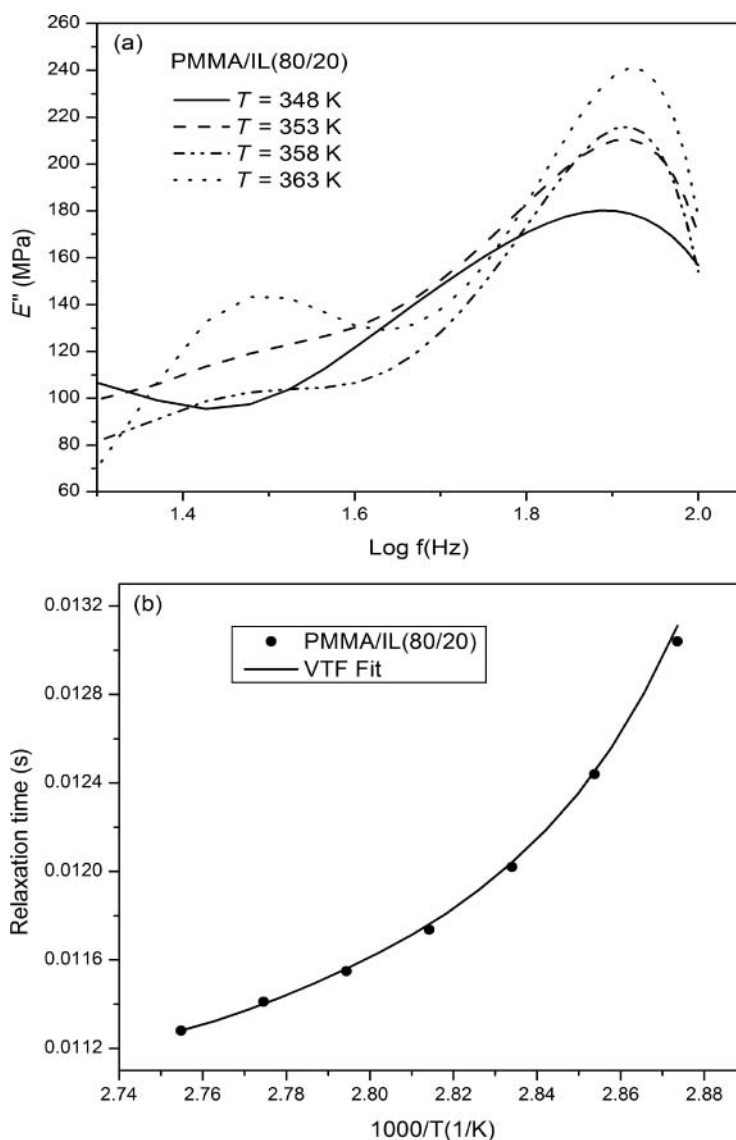


Figure 5. (a) Frequency dependence of loss modulus at several temperatures in the glass transition region for PMMA/BMIMPF₆ (80/20) blend sample; (b) temperature dependency of relaxation time for PMMA/BMIMPF₆ (80/20) blend system.

reported.^[45] Thus, it can be inferred that these two distinct α -relaxation peaks observed at high temperatures cannot be interpreted based on phase separation. On the other hand, this observed phenomenon has been reported for miscible blend systems with a large dynamical asymmetry, that is, T_g difference. A $\Delta T_g \approx 200$ K, between PMMA and BMIMPF₆, can lead to intrinsic mobility differences between the two component chains. When this blend system is subjected to external stress, self-concentration effects together with concentration fluctuations and intrinsic mobility differences can finally lead to two distinct peaks for the α -relaxation, the so-called dynamic heterogeneity, in the case of a polymer/IL mixture with low amounts of IL (20 wt%).

The characteristic time for local segmental relaxation was calculated by the frequency corresponding to the loss function peak, $\tau \approx 1/f_{\max}$. The temperature dependence of the relaxation time for the 20/80 sample was calculated and is plotted in Fig. 5(b). The curve through the data represents the best fit to the Vogel–Fulcher–Tamman (VFT) equation:

$$\tau = \tau_0 \exp\left(\frac{B}{T - T_\infty}\right), \quad (2)$$

where B is the material parameter defining the relaxation activation energy and T_∞ is the Vogel temperature representing the point, where the relaxation time diverges to infinity. The pre-exponential τ_0 corresponds to the relaxation time at infinite temperature.

Solid-state NMR has proven to be a powerful tool in studying molecular and ionic dynamics in polymer electrolytes. As an element-selective method able to probe local interactions and chemical environments, the NMR lineshape and spin–lattice relaxation measurements as a function of temperature can provide an effective method to probe molecular and ionic mobility on the time scale of microseconds to seconds.^[46–52] Such information is helpful to have a deeper understanding of ion mobility and conductivity in polymer electrolytes.

Figure 6 shows the ⁷Li spin–lattice relaxation rate ($1/T_1$) as a function of temperature for the PMMA/BMIMPF₆ solid-state electrolyte at two different compositions. Here, it was necessary to use a two-component fit, thereby determining two T_1 values for ⁷Li for both compositions at all temperatures. This suggests that there are at least two Li ion sites for all temperatures. The presence of two different Li ion sites can be ascribed to the two-component structures of these electrolytes. Both the PMMA and the IL can interact and dissociate the lithium salt, resulting in two different Li ion sites. Another interesting feature in this figure is the high and almost temperature-independent relaxation rates for both systems. It is surprising to see the apparently flat curve of the ($1/T_1$) relaxation rates over a wide temperature range shown in Fig. 6. This is in contrast to the well-defined maximum normally observed in liquid electrolytes with single exponential magnetisation decays. However, it should be mentioned that the relaxation rate value shown here is within the range of maximum values previously reported for both liquid or gel electrolytes.^[1,51,52] We note, however, there is a slight reduction in relaxation rate with decreasing temperature. This decrease in the relaxation rate ($1/T_1$) indicates a decrease in mobility. This would indicate a stronger interaction between Li ions and IL and/or polymer as temperature decreases; therefore hindering the motion and relaxation of molecules as temperature decreased. It seems that ⁷Li spin–lattice relaxation measurements in a wider temperature range would be helpful for a conclusive analysis.

Fractions of both “fast” and “slow” Li ions, lithium ions with shorter and longer spin–lattice relaxation rates, respectively, as a function of temperature for two different blend compositions are shown in Fig. 7. The fraction of fast Li ions in the 50/50 blend

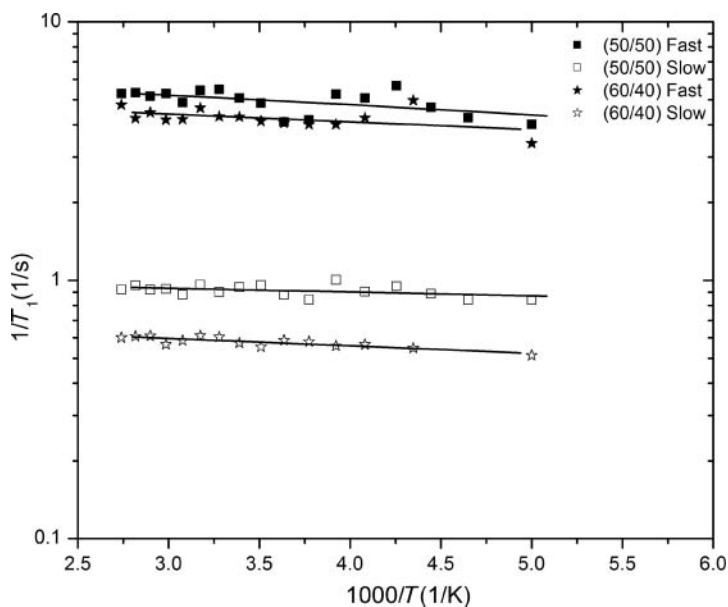


Figure 6. ^7Li spin-lattice relaxation time as a function of temperature for PMMA/BMIMPF₆ solid-state electrolyte at two different compositions.

system was higher than that of the other system (60/40). More importantly, the fraction of fast Li ions in both of the blend systems increased with increasing temperature, indicating that more and more of the short T_1 Li ions were changing their environment compared with the longer T_1 lithium ions. It should also be mentioned that the increase in fast Li ion fraction with increasing temperature was more noticeable for the 50/50 blend system. These

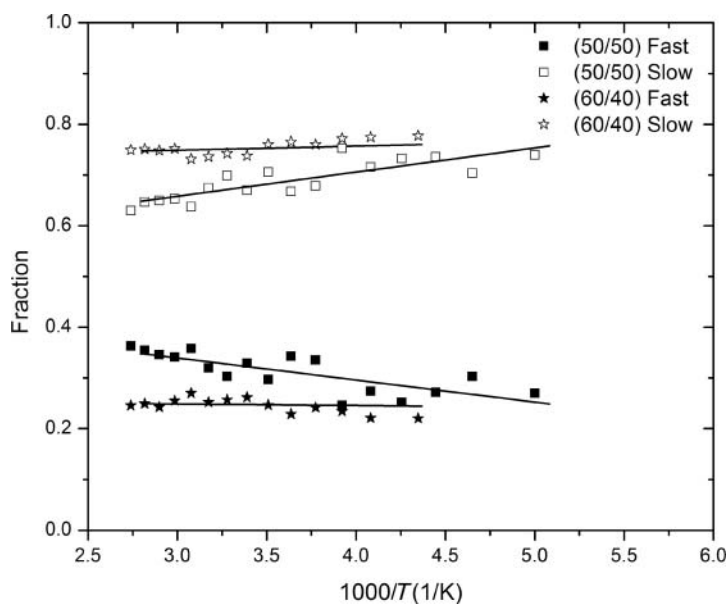


Figure 7. The fraction of both “fast” and “slow” Li-ions as a function of temperature for PMMA/BMIMPF₆ blend system in two different compositions.

Table 2
Ionic conductivity of PMMA/BMIMPF₆/LiClO₄ blend systems as a function of IL content at room temperature

BMIMPF ₆ content (wt%)	σ (S/cm)
0	0.707×10^{-9}
20	0.199×10^{-7}
40	0.125×10^{-5}
50	0.84×10^{-5}

results indicate that the Li ions were changing their preferred interactions with increasing temperature.

Ionic conductivity values of the electrolyte membranes as a function of IL content at room temperature are presented in Table 2. It can be seen that the conductivity increased steadily with increasing the amount of IL. Figure 8 shows the ionic conductivity as a function of temperature for these solid-state electrolyte membranes. Although a Vogel–Tamman–Fulcher (VTF) kind of temperature dependency was expected, the linear pattern is suggestive of an Arrhenius relationship for all the samples. In such a molecularly mixed mixture, each BMIMPF₆ molecule is nearly surrounded by PMMA segments. At these studied mixture compositions that are well above the overlap weight fraction of BMIMPF₆ in the PMMA matrix, the ionic conductivity of the membranes increased with temperature. This increase was more noticeable as the amount of IL increased. However, it should be mentioned that at almost all the studied temperature range, the PMMA segments were far below their glass transition temperatures and therefore were structurally solid. Therefore, lithium ion conduction in this mixture can be mainly attributed to confined IL

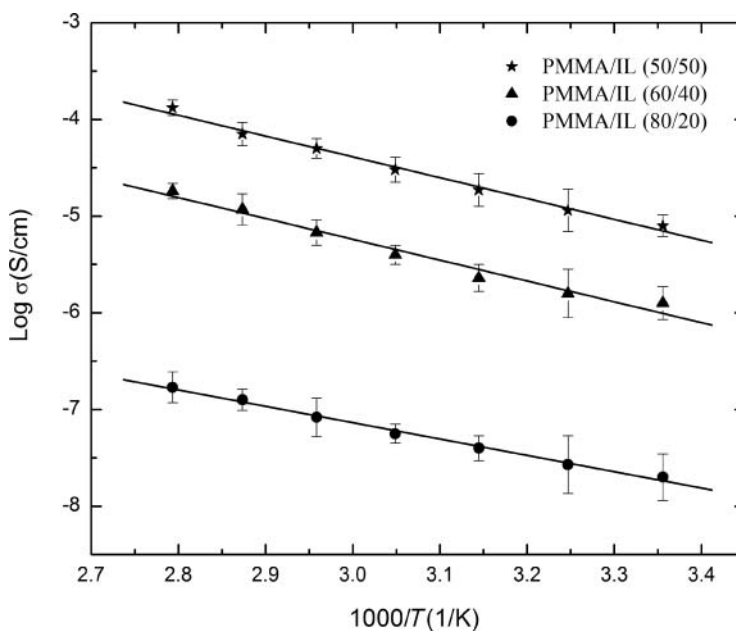


Figure 8. Relationship between ionic conductivity and temperature for PMMA/BMIMPF₆/LiClO₄ blend systems.

molecules in the rigid PMMA matrix. The role of the rigid PMMA matrix chains was mostly confining the IL molecules, which can result in acceleration of confined molecules movements and facilitating lithium ion conduction. Furthermore, in all of the plots in Fig. 8, there was a smooth variation of the conductivity in the entire temperature range of investigation confirming the thermal stability of the membranes.

Conclusions

Miscible mixtures of PMMA/BMIMPF₆ were prepared by the solution casting method. Differential scanning calorimetry and dynamical mechanical thermal analysis confirmed the formation of miscible and homogeneous polymer/IL systems. Solid-state NMR measurements showed a high and almost temperature-independent spin–lattice relaxation rate ($1/T_1$) for the investigated PMMA/BMIMPF₆ blends. This result reflects a basic difference between the spin–lattice relaxation of these systems and that of conventional gel polymer electrolytes in which a maximum is normally observed. A slight decrease in relaxation rate with decreasing temperature indicates stronger interactions between the components in the electrolyte systems at lower temperatures. From AC impedance spectroscopy, the ionic conductivity increased by increasing the amount of IL in the blend, suggesting that confined IL molecules were mainly responsible for the lithium ion conduction in this mixture.

Acknowledgments

The authors would like to thank Claude Lemaire in the Spectroscopy and Macroimaging Laboratory of the Physics and Astronomy Department of the University of Waterloo. His great help is highly appreciated.

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